

Virginia Tech
Department of Economics
Fall 2026

ECON 4074: Labor Economics

Instructor: Brianna Felegi, Ph.D. (bfelegi@vt.edu)

Teaching Assistant: Reza Tavakoli (rtavakoli@vt.edu)

Class Time: 3:30 to 4:45pm on Tuesdays and Thursdays

Class Location: ROB103

Instructor Office Hours: Tuesdays and Thursdays in Pamplin 3122
8:30-10:30am, 2:00-3:00pm

TA Office Hours: Wednesdays in Pamplin 3053
9:00-10:30am

Course Description and Objectives

This course is an introduction to the field of labor economics with an emphasis on policy applications. We will cover a variety of topics including labor supply, labor demand, compensating wage differentials, education, immigration, and discrimination. From this course, students will develop an understanding of how economic research is used to analyze U.S. labor markets, how such research influences policy debates, and how to apply economic tools to answer their own labor-related questions.

Prerequisites

ECON 2005 (Principles of Economics – Micro), ECON 2006 (Principles of Economics – Macro), ECON3254 (Analysis of Economic Data) or ECON4304 (Econometric Methods)

Course Materials

1. “Labor Economics”, 2024, George Borjas
2. Stata, A six-month student copy of the I/C version is sufficient ([link](#))

Statistical Software

The class will use the Stata statistical software package. Stata is a fast and versatile program that was written by economists, so it is more intuitive for people in our field. Stata is the program of choice for applied micro economists.

You can purchase a six-month student copy of Stata for \$48 at the following link:

<https://www.stata.com/order/new/edu/profplus/student-pricing/>

Stata is a required resource. This software will not run on Chromebooks. The details of computer requirements for Stata can be found here:

<https://www.stata.com/products/compatible-operating-systems/>

I have programmed in Stata for years so if you have any question, please let me know. One note – to help you with your programming, I need to see the code. Therefore, copy the code you are working and email me the text. Too many people come to my office and say “I typed what was in the handout and it did not work” then don’t show me their code. I need to see what you are doing before I can help.

Course Components (and grading weight)

A. Participation (5%)

Students should be prepared to contribute to class discussions (answering questions, asking clarifying questions, going to office hours, pointing out mistakes on board, providing feedback to peers during group presentations). Throughout the semester, I will use notecards to cold-call students. You can earn participation points by attempting to answer the questions I pose.

B. Homework Assignments (15%)

Homework will involve solving problems and answering questions related to material we covered in class, as well as performing econometric analysis in Stata. You may use the textbook, personal notes, or other reference material to do the homework. You may and are encouraged to work with your classmates. However, everyone must turn in their own work, perform their own Stata analysis, and write the answers in their own words. If homework is identical to a classmates’, you may not receive credit for the answer. There are four problem sets in total.

Assignments and due dates will be posted in Canvas. Homework is to be turned in by 11:59pm on the due date. You can turn in a physical copy to me in class or upload a submission in Canvas. For Stata portions of the homework, you must submit your code and output. **You cannot receive more than a 70% if you do not include the required do-file and log-file.** 10 points will be taken off for each day past the due date that the homework is turned in. Once the homework solutions are posted, you cannot receive above a 40%.

C. Student Discussion Days (20%)

Four meetings over the semester are allocated as student discussion days. These days are intended to serve three purposes: (1) provide the opportunity for students to dive deeper into specific topics, (2) help students connect what they are learning in the classroom to current policy debates, and (3) help students develop the skill of talking and thinking about economics in a practical manner.

For these meetings, presenting students are expected to prepare by carefully reading a news article of their choice related to prior weeks’ lectures. Students are also expected to identify and discuss one academic economics paper related to the news article. Students that are not submitting items for discussion on these days are expected to provide three questions for each of the presenters and attend the session itself. Deadlines for submitting the materials and

questions, as well as grading rubrics, will be posted on Canvas. I will assign students to specific student discussion days at the end of the second week of the semester.

Attendance is required on student discussion days. They will take place on the following dates:

- Thursday, September 17th
- Thursday, October 8th
- Thursday, October 29th
- Tuesday, December 1st

D. Data Dashboard Assignment (20%)

The data dashboard assignment consists of you (and up to two other classmates) presenting the findings of an assigned national statistics report related to a topic taught in the course. Each group is expected to read and interpret the release, then rebuild at least four data visualizations. The data visualizations are expected to be transformed in a meaningful way beyond what appears in the original report. Students will present their findings to the class and submit a 3-4-page memo describing their results.

There will be five components of the dashboard assignment that will be graded.

1. Data Sourcing and Code (10%) Each group will submit the data and code used to complete the assignment to assess replicability and accuracy of the report.
2. Data Visualizations (20%) Each group will be assessed on how they transformed the figures or tables from the original report, the readability of the new visualization, and the accuracy of the conclusions made from the from them.
3. Class Presentation (30%) Each group will present their data dashboard during a 10-minute presentation in class. Each group is required to discuss the main findings in the report, the data sources used to update the data visualizations and connect the findings to relevant policy debates.
4. Written Report (30%) Each group is required to hand in a 3-4-page report, that summarizes the main findings from the assigned report, discusses how the group transformed the data visualizations, what data was used, and how the data visualizations inform policy debates today.
5. Participation (10%) Each group member will fill out a report that grades each members participation. Each member will receive the average grade assigned by their group members.

E. Research Project (40%)

You will complete a research report that answers a labor economics research question using the theoretical concepts and statistical techniques developed in this course. The project will require you to develop a labor economic research question, find data to answer the question and perform the statistical analysis necessary to answer the question.

There will be five components of the project that will be graded.

1. Project Proposal (20%) – Each student will submit three potential research questions they wish to answer for their project. This assignment requires students to explain why the research questions are interesting, what potential data sources they can use, discuss relevant literature on the topic, and how the question is related to the course.
2. Data Description and Summary Statistics (20%) – Each student will submit a 2-3-page document describing the dataset they have chosen (i.e., structure, number of observations, summary statistics of key variables of interest, etc.).
3. Regression Specification and Preliminary Results (20%) – Each student will submit a 2-3-page document describing the chosen regression model to answer the research question as well as the preliminary results.
4. Literature Review, Theory, Limitations, and Policy Implications (20%) – Each student will submit a 3-4-page document that includes a literature review, discusses the relationship between your research question and economic theory, discusses the limitations of your methodology, and provides policy implications from your findings.
5. Final Report (20%) – Each student is required to hand in a 10-15-page report, that provides background information, your research question, discussion of your model, data used, presentation of your results, their implications as well as their limitations. Students are expected to incorporate the feedback from the prior assignments.

Grading rubric and an example of a completed project will be provided during the semester.

In summary, final grades for the course will be based on the following:

Participation	5%
Homework Assignments	15%
Student Discussion Days	20%
Data Dashboard Assignment	20%
Research Project	40%

Letter grades will be assigned as follows:

A	93 - 100
A-	90 – 92.99
B+	87 – 89.99
B	83 – 86.99
B-	80 – 82.99
C+	77 – 79.99
C	73 – 76.99
C-	70 – 72.99
D+	67 – 69.99
D	63 – 66.99
D-	60 – 62.99
F	0 – 59.99

Grading Questions or Disputes

I want you to understand how you are evaluated in this class. If you have any questions about your grade or feedback on course work, please email me to schedule a time to meet. I kindly ask that you wait at least 24 hours after work has been returned before requesting an appointment. This is to give you time to more thoroughly consider questions you have about your work.

Canvas

Our website within Canvas (<https://canvas.vt.edu/>) will have the slides class, assignment information and more. Please check the website regularly for class updates.

Academic Honesty

One of the goals of this course is to prepare you to do well in advanced economics courses, so it is essential that you develop a solid understanding of the material. To that end, while collaboration on problem sets is encouraged in this course, copying is not. Students are expected to understand and abide by the University's honor code (<https://honorsystem.vt.edu/>).

Undergraduate Academic Integrity Statement

The Undergraduate Honor Code pledge that each member of the university community agrees to abide by states:

“As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do.”

Students enrolled in this course are responsible for abiding by the Honor Code. A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Students are strongly discouraged from misusing sites such as Chegg and CourseHero, as well as misusing ChatGPT and other Generative Artificial Intelligence. Students are strongly encouraged to consult their faculty members regarding the use of such outside materials as the misuse of these sources may constitute a violation of the Honor Code. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code.

Academic Accommodations

Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to disability, including but not limited to ADHD, chronic or temporary medical conditions, deaf or hard of hearing, learning disability, mental health, or vision impairment, please contact the Services for Students with Disabilities (SSD) office (540-231-3788, ssd@vt.edu, or visit ssd.vt.edu). If you have an SSD accommodation letter, please meet with me privately during office hours as early in the semester as possible to deliver your letter and discuss your accommodations. You must give me reasonable notice to implement your accommodations, which is generally 5 business days and 10 business days for final exams.

If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together we'll develop strategies to meet both your needs and the requirements of the course.

Basic Needs Statement

For any student who has difficulty affording groceries, accessing sufficient food to eat every day, or who lacks a safe and stable place to live, and if you believe this may affect your performance in this course, you are urged to contact the Dean of Students office for support at 540-231-3787 or complete an [interest form](#) to participate in The Market at Virginia Tech. The Dean of Students, through The Market at Virginia Tech, offers food options and other resources. There is also a [Student Emergency Fund program](#). If you are comfortable in doing so, please notify me or your departmental advisor of your situation. This will enable them to provide any resources they have access to.

Lauren's Promise I will listen and believe you if someone is threatening or harassing you.

Lauren McCluskey, a 21-year-old student at the University of Utah and the daughter of economist Jill McCluskey, was murdered on October 22nd, 2018, by a man she briefly dated. I am committed to doing what I can to make sure this does not happen again.

If you are in immediate danger, call 911. If you are experiencing sexual assault, domestic violence, stalking, or harassment you can report it to me. You should be aware that I am a mandatory reporter, which means that I am required to report these instances to the university's Title IX Coordinator or Deputy Title IX Coordinator to investigate. **You can also learn about your options (both on and off-campus as well as confidential or not) at this [link](https://ucc.vt.edu/).** If you need support for your mental or emotional health for any reason, you can talk to me. You can also find helpful resources at <https://ucc.vt.edu/>

Tentative Course Topic Outline – Chapters outlined for Labor Economics by Borjas

Dates	Lecture Topics	Assignments and Readings
Week I: Aug. 25th & Aug. 27th	Introduction to Course Review of Empirical Tools	Chap. 1- Introduction
Week II: Sep. 1st & Sep. 3rd	Review of Empirical Tools Labor Supply	Chap. 2 – Labor Supply
Week III: Sep. 8th & Sep. 10th	Labor Supply	Sep. 13 th – Discussion Group 1 Due
Week IV: Sep. 15th & Sep. 17th	Labor Supply Student Discussion Day 1	Sep. 20 th – Problem Set #1 Due
Week V: Sep. 22nd & Sep. 24th	Labor Demand	Chap. 3 – Labor Demand Sep. 27 th – Project Proposal Due
Week VI: Sep. 29th & Oct. 1st	Labor Market Equilibrium	Chap. 4 – Labor Market Equilibrium
Week VII: Oct. 6th & Oct. 8th	Labor Market Equilibrium Student Discussion Day 2	Oct. 4 th – Discussion Group 2 Due Oct. 11 th – Problem Set #2 Due
Week VIII: Oct. 13th & Oct. 15th	Compensating Wage Differentials	Chap. 5 – Compensating Wage Differentials Oct. 18 th – Descriptive Statistics Due

Week IX: Oct. 20th & Oct. 22nd	Education	Chap. 6 – Education Oct. 25 th – Discussion Group 3 Due
Week X: Oct. 27 th & Oct. 29 th	Education Student Discussion Day 3	Nov. 1 st – Problem Set #3 Due
Week XI: Nov. 3 rd & Nov. 5 th	Labor Mobility	Chap. 8 – Labor Mobility Nov. 8 th – Preliminary Results Due
Week XII: Nov. 10 th & Nov. 12 th	Labor Mobility	
Week XIII: Nov. 17 th & Nov. 19 th	Labor Market Discrimination	Chap. 9 – Labor Market Discrimination
Week XIV: Nov. 24 th & Nov. 26 th	No Class – Thanksgiving Break	Nov. 29 th – Discussion Group 4 Due
Week XV: Dec. 1st & Dec. 3rd	Student Discussion Day 4	Dec. 1 st – Report Draft Due Dec. 6 th – Problem Set #4 Due
Week XVI: Dec. 8th	Research Project Day	December 16 th – Final Report Due